

Conceptual Question #1

Using a little bit of algebra, prove that Equation (4.2) is equivalent to Equation (4.3). In other words, show that the logistic function representation and logit representation for the logistic regression model are equivalent.

Starting with the logistic regression model,

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Let

$$z = \beta_0 + \beta_1 X.$$

Then,

$$p(X) = \frac{e^z}{1 + e^z}.$$

Now compute $(1 - p(X))$:

$$1 - p(X) = 1 - \frac{e^z}{1 + e^z}$$

$$1 - p(X) = \frac{1 + e^z - e^z}{1 + e^z}$$

$$1 - p(X) = \frac{1}{1 + e^z}.$$

Next, form the odds ratio:

$$\frac{p(X)}{1 - p(X)} = \frac{\frac{e^z}{1 + e^z}}{\frac{1}{1 + e^z}}$$

$$\frac{p(X)}{1 - p(X)} = e^z.$$

Substituting back,

$$z = \beta_0 + \beta_1 X,$$

gives

$$\frac{p(X)}{1-p(X)} = e^{\beta_0 + \beta_1 X}.$$

This is Equation (4.3), so Equation (4.2) and Equation (4.3) are algebraically equivalent.

Taking the natural logarithm of both sides yields

$$\log\left(\frac{p(X)}{1-p(X)}\right) = \beta_0 + \beta_1 X.$$

Therefore, the logistic function representation, odds representation, and logit representation are equivalent ways of expressing the same logistic regression model.

Applied Question #13

The objective of this exercise is to evaluate several classification methods using the Weekly data set from ISLP. The response variable is Direction, which indicates whether the market went Up or Down. The analysis will include logistic regression, LDA, QDA, Naive Bayes, and KNN, followed by a comparison of prediction performance.

```
In [1]: # Import required libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import statsmodels.api as sm

from ISLP import load_data
from ISLP.models import ModelSpec as MS
from ISLP import confusion_table

from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
from sklearn.discriminant_analysis import QuadraticDiscriminantAnalysis as QDA
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import confusion_matrix, accuracy_score

# Load Weekly dataset
```

```
Weekly = load_data("Weekly")
# Display first few observations
Weekly.head()
```

Out [1]:

	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today	Direction
0	1990	0.816	1.572	-3.936	-0.229	-3.484	0.154976	-0.270	Down
1	1990	-0.270	0.816	1.572	-3.936	-0.229	0.148574	-2.576	Down
2	1990	-2.576	-0.270	0.816	1.572	-3.936	0.159837	3.514	Up
3	1990	3.514	-2.576	-0.270	0.816	1.572	0.161630	0.712	Up
4	1990	0.712	3.514	-2.576	-0.270	0.816	0.153728	1.178	Up

Part (a)

Produce some numerical and graphical summaries of the Weekly data. Do there appear to be any patterns? Aside from the strong increase in trading volume over time, there is little evidence of meaningful relationships among the lagged returns, suggesting limited predictability of market direction from past weekly returns.

```
In [2]: # Basic information about the dataset

Weekly.info()

print("\n")
print("Dataset Shape:", Weekly.shape)

print("\nSummary Statistics:")
display(Weekly.describe(include='all'))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1089 entries, 0 to 1088
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Year        1089 non-null   int64
1   Lag1        1089 non-null   float64
2   Lag2        1089 non-null   float64
3   Lag3        1089 non-null   float64
4   Lag4        1089 non-null   float64
5   Lag5        1089 non-null   float64
6   Volume      1089 non-null   float64
7   Today       1089 non-null   float64
8   Direction   1089 non-null   category
dtypes: category(1), float64(7), int64(1)
memory usage: 69.4 KB
```

Dataset Shape: (1089, 9)

Summary Statistics:

	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today	Dire
count	1089.000000	1089.000000	1089.000000	1089.000000	1089.000000	1089.000000	1089.000000	1089.000000	
unique	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
top	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
freq	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
mean	2000.048669	0.150585	0.151079	0.147205	0.145818	0.139893	1.574618	0.149899	
std	6.033182	2.357013	2.357254	2.360502	2.360279	2.361285	1.686636	2.356927	
min	1990.000000	-18.195000	-18.195000	-18.195000	-18.195000	-18.195000	0.087465	-18.195000	
25%	1995.000000	-1.154000	-1.154000	-1.158000	-1.158000	-1.166000	0.332022	-1.154000	
50%	2000.000000	0.241000	0.241000	0.241000	0.238000	0.234000	1.002680	0.241000	
75%	2005.000000	1.405000	1.409000	1.409000	1.409000	1.405000	2.053727	1.405000	
max	2010.000000	12.026000	12.026000	12.026000	12.026000	12.026000	9.328214	12.026000	

```
In [3]: # Correlation matrix for numerical variables

Weekly_numeric = Weekly.select_dtypes(include=[np.number])

corr_matrix = Weekly_numeric.corr()

corr_matrix
```

Out [3]:

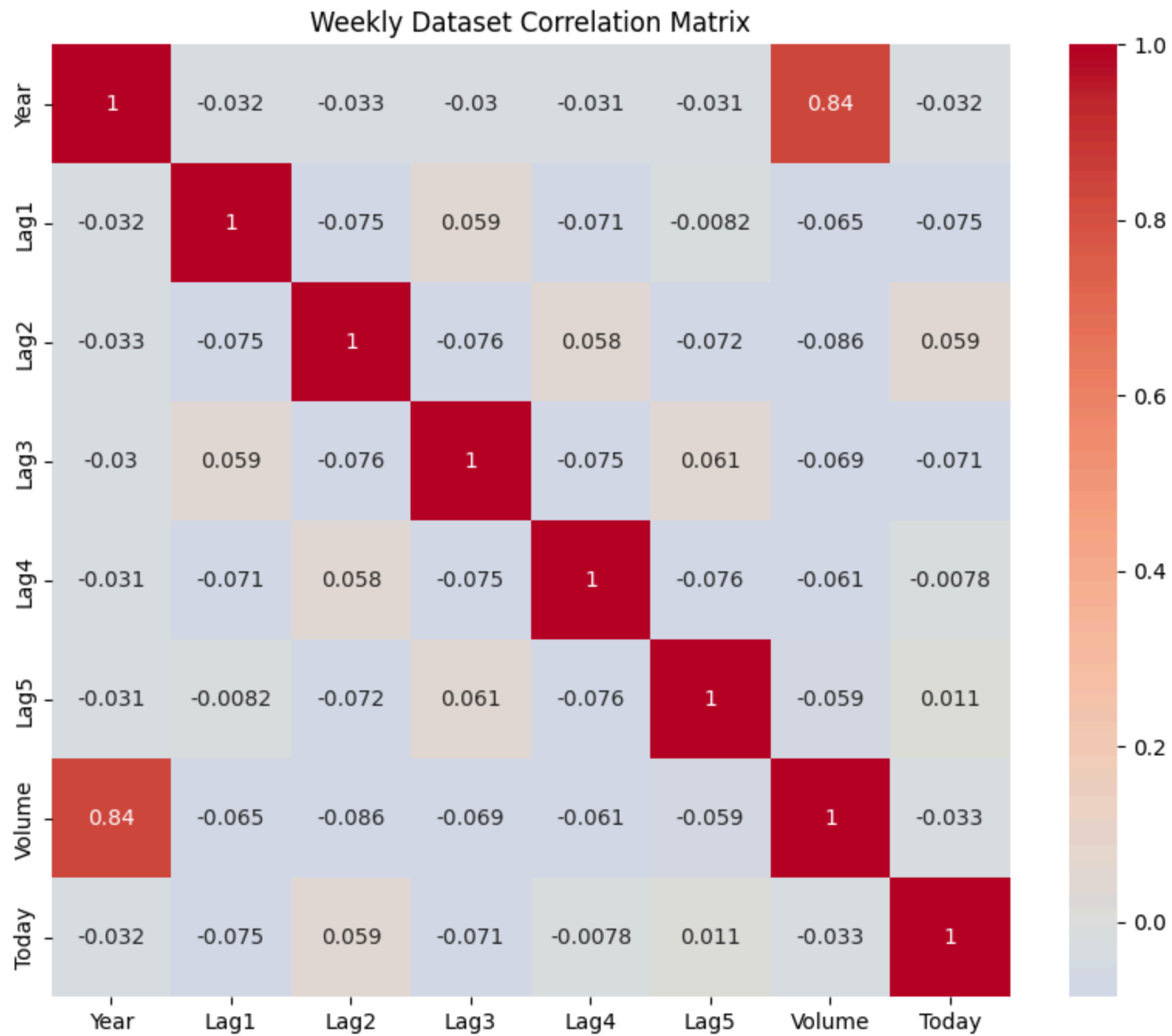
	Year	Lag1	Lag2	Lag3	Lag4	Lag5	Volume	Today
Year	1.000000	-0.032289	-0.033390	-0.030006	-0.031128	-0.030519	0.841942	-0.032460
Lag1	-0.032289	1.000000	-0.074853	0.058636	-0.071274	-0.008183	-0.064951	-0.075032
Lag2	-0.033390	-0.074853	1.000000	-0.075721	0.058382	-0.072499	-0.085513	0.059167
Lag3	-0.030006	0.058636	-0.075721	1.000000	-0.075396	0.060657	-0.069288	-0.071244
Lag4	-0.031128	-0.071274	0.058382	-0.075396	1.000000	-0.075675	-0.061075	-0.007826
Lag5	-0.030519	-0.008183	-0.072499	0.060657	-0.075675	1.000000	-0.058517	0.011013
Volume	0.841942	-0.064951	-0.085513	-0.069288	-0.061075	-0.058517	1.000000	-0.033078
Today	-0.032460	-0.075032	0.059167	-0.071244	-0.007826	0.011013	-0.033078	1.000000

In [4]: *# Visualize correlations*

```
import seaborn as sns

plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix,
            annot=True,
            cmap="coolwarm",
            center=0)

plt.title("Weekly Dataset Correlation Matrix")
plt.show()
```

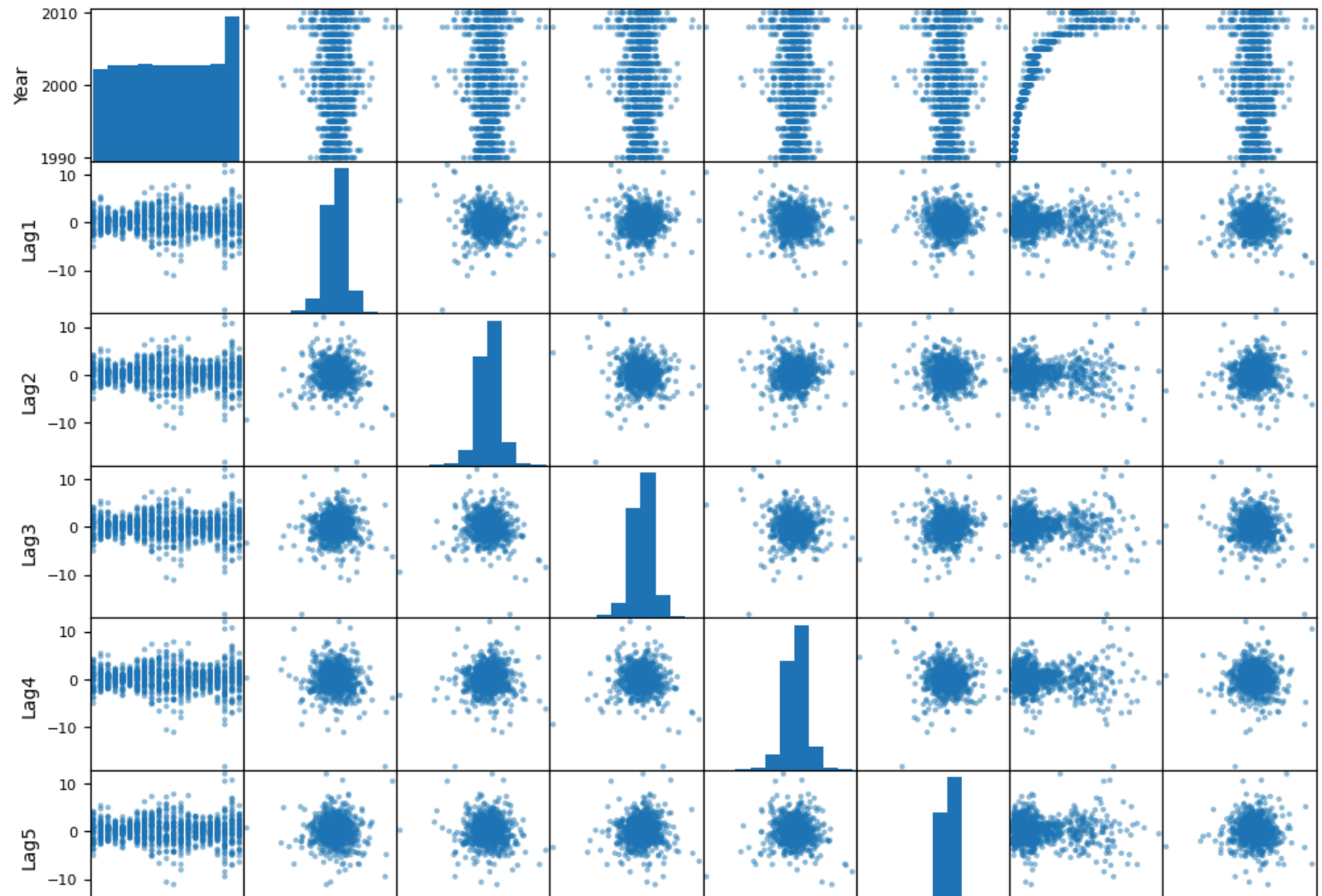


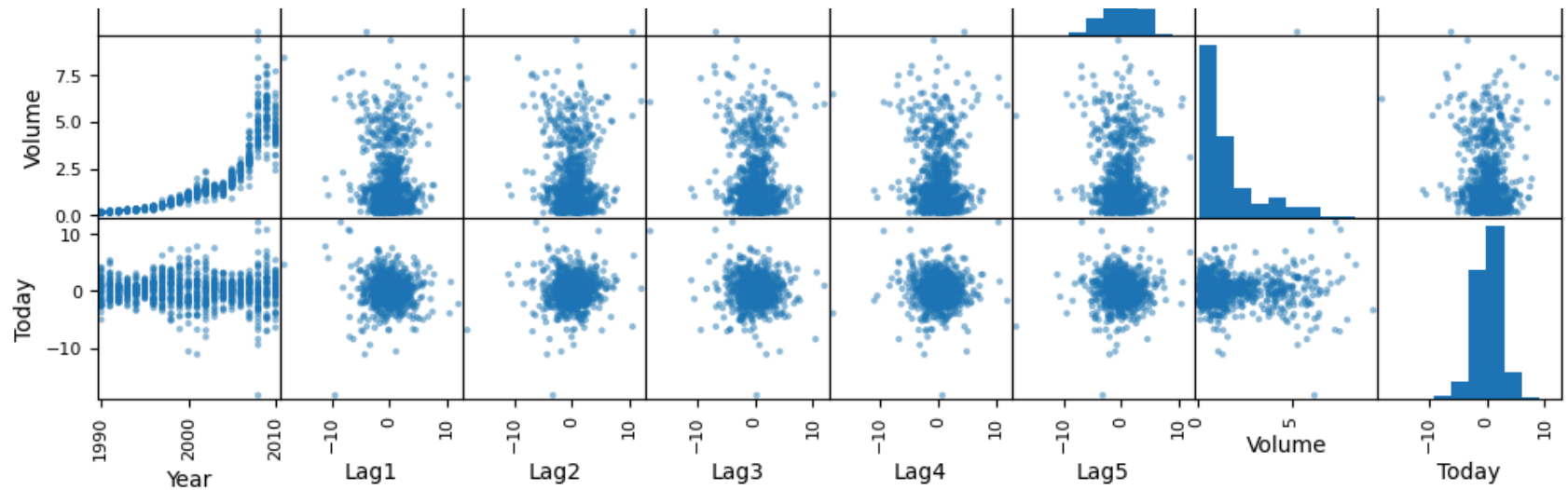
```
In [5]: # Scatterplot matrix

pd.plotting.scatter_matrix(
    Weekly_numeric,
    figsize=(12,12),
    diagonal='hist'
)

plt.suptitle("Scatterplot Matrix of Weekly Data", y=1.02)
plt.show()
```


Scatterplot Matrix of Weekly Data





Interpretation

The Weekly dataset contains 1,089 observations and 9 variables. The response variable, **Direction**, has two categories (Up and Down), with Up occurring slightly more frequently (605 observations).

From the correlation matrix and heatmap, most predictor variables (Lag1–Lag5 and Today) exhibit very weak correlations with one another and with the response-related variables. The only notable relationship is between **Year** and **Volume**, which have a strong positive correlation of approximately **0.84**. This suggests that trading volume has generally increased over time.

The scatterplot matrix confirms these findings. There are no strong linear relationships among the lag variables, and the observations appear widely dispersed. However, the relationship between Year and Volume shows a clear upward trend, indicating that market trading activity increased substantially during the period covered by the data.

Overall, there is little evidence of strong associations among the lagged return variables, suggesting that predicting market direction based solely on past weekly returns may be challenging. The strongest pattern observed is the steady increase in trading volume over time.

Part (b)

Use the full data set to perform a logistic regression with Direction as the response and Lag1, Lag2, Lag3, Lag4, and Lag5 plus Volume as predictors. Use the summary function to print the results. Do any of the predictors appear to be statistically significant? If so, which ones?

```
In [6]: # Part (b): Logistic Regression using all predictors

Weekly["Direction_binary"] = (Weekly["Direction"] == "Up").astype(int)

X = Weekly[["Lag1", "Lag2", "Lag3", "Lag4", "Lag5", "Volume"]]
X = sm.add_constant(X)

y = Weekly["Direction_binary"]

logit_model = sm.Logit(y, X).fit()

print(logit_model.summary())
```

Optimization terminated successfully.

Current function value: 0.682441

Iterations 4

Logit Regression Results

```
=====
Dep. Variable:    Direction_binary    No. Observations:    1089
Model:                Logit    Df Residuals:    1082
Method:                MLE    Df Model:    6
Date:                Sat, 30 May 2026    Pseudo R-squ.:    0.006580
Time:                16:46:08    Log-Likelihood:    -743.18
converged:                True    LL-Null:    -748.10
Covariance Type:    nonrobust    LLR p-value:    0.1313
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	0.2669	0.086	3.106	0.002	0.098	0.435
Lag1	-0.0413	0.026	-1.563	0.118	-0.093	0.010
Lag2	0.0584	0.027	2.175	0.030	0.006	0.111
Lag3	-0.0161	0.027	-0.602	0.547	-0.068	0.036
Lag4	-0.0278	0.026	-1.050	0.294	-0.080	0.024
Lag5	-0.0145	0.026	-0.549	0.583	-0.066	0.037
Volume	-0.0227	0.037	-0.616	0.538	-0.095	0.050

Interpretation

The logistic regression model was fit using Direction as the response variable and Lag1, Lag2, Lag3, Lag4, Lag5, and Volume as predictors.

To determine statistical significance, we examine the p-values associated with each predictor. At the 5% significance level ($\alpha = 0.05$), only **Lag2** is statistically significant with a p-value of approximately **0.030**.

The coefficient for Lag2 is positive (0.0584), indicating that higher returns two weeks ago are associated with an increased probability that the market will move Up in the current week.

All other predictors (Lag1, Lag3, Lag4, Lag5, and Volume) have p-values greater than 0.05 and therefore do not appear to be statistically significant predictors of market direction in this model.

Thus, based on the logistic regression results, **Lag2 is the only predictor that shows a statistically significant relationship with Direction.**

Part (c)

Compute the confusion matrix and overall fraction of correct predictions. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.

```
In [7]: # Part (c): Confusion Matrix and Accuracy

# Predicted probabilities
pred_probs = logit_model.predict(X)

# Convert probabilities to class predictions
pred_direction = np.where(pred_probs > 0.5, "Up", "Down")

# Confusion matrix
cm = confusion_matrix(Weekly["Direction"], pred_direction)

print("Confusion Matrix:")
print(cm)
```

```
# Accuracy
accuracy = accuracy_score(Weekly["Direction"], pred_direction)

print("\nAccuracy:", round(accuracy, 4))

# Display with labels
cm_df = pd.DataFrame(
    cm,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_df)
```

Confusion Matrix:

```
[[ 54 430]
 [ 48 557]]
```

Accuracy: 0.5611

	Predicted Down	Predicted Up
Actual Down	54	430
Actual Up	48	557

Interpretation

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	54	430
Actual Up	48	557

The overall classification accuracy is approximately **56.1%**.

The model correctly classified 54 Down weeks and 557 Up weeks, for a total of 611 correct predictions out of 1,089 observations. However, the model performs much better at predicting Up weeks than Down weeks.

Of the 484 actual Down weeks, only 54 were correctly classified, while 430 were incorrectly classified as Up. In contrast, the model correctly classified 557 of the 605 actual Up weeks.

This indicates that the logistic regression model has a strong tendency to predict **Up** regardless of the actual market direction. As a result, it achieves moderate overall accuracy because Up weeks occur slightly more frequently than Down weeks, but it performs poorly at identifying Down weeks.

Part (d)

Now fit the logistic regression model using a training data period from 1990–2008 and test the model on observations from 2009–2010. Use Lag2 as the only predictor. Compute the confusion matrix and overall accuracy.

```
In [8]: # Part (d): Train/Test Split using Lag2 only

# Training observations: years before 2009
train = Weekly["Year"] < 2009

# Training and test sets
train_data = Weekly[train]
test_data = Weekly[~train]

# Predictor
X_train = sm.add_constant(train_data[["Lag2"]])
X_test = sm.add_constant(test_data[["Lag2"]])

# Response
y_train = (train_data["Direction"] == "Up").astype(int)

# Fit logistic regression
logit_lag2 = sm.Logit(y_train, X_train).fit()

# Predict on test set
test_probs = logit_lag2.predict(X_test)

test_pred = np.where(test_probs > 0.5, "Up", "Down")

# Confusion matrix
cm_test = confusion_matrix(test_data["Direction"], test_pred)
```

```

print("Confusion Matrix:")
print(cm_test)

# Accuracy
accuracy_test = accuracy_score(test_data["Direction"], test_pred)

print("\nTest Accuracy:", round(accuracy_test, 4))

# Labeled confusion matrix
cm_test_df = pd.DataFrame(
    cm_test,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_test_df)

```

Optimization terminated successfully.
 Current function value: 0.685555
 Iterations 4

Confusion Matrix:
 [[9 34]
 [5 56]]

Test Accuracy: 0.625

	Predicted Down	Predicted Up
Actual Down	9	34
Actual Up	5	56

Interpretation

The logistic regression model was trained using observations from 1990–2008 and tested on observations from 2009–2010, with Lag2 as the only predictor.

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	9	34
Actual Up	5	56

The overall test accuracy is **62.5%**.

The model correctly classified 65 of the 104 test observations. It correctly identified 56 of the 61 Up weeks but only 9 of the 43 Down weeks.

Similar to the previous model, logistic regression tends to predict Up more frequently than Down. However, the out-of-sample accuracy of 62.5% is noticeably better than the 56.1% obtained when fitting and evaluating the model on the entire dataset.

These results suggest that Lag2 contains useful predictive information about market direction and provides modest predictive power for forecasting future market movements.

Part (e)

Repeat part (d) using Linear Discriminant Analysis (LDA). Compute the confusion matrix and overall accuracy.

```
In [9]: # Part (e): LDA using Lag2

# Training data
X_train = train_data[["Lag2"]]
X_test = test_data[["Lag2"]]

y_train = train_data["Direction"]

# Fit LDA
lda_model = LDA()
lda_model.fit(X_train, y_train)

# Predictions
lda_pred = lda_model.predict(X_test)

# Confusion matrix
cm_lda = confusion_matrix(test_data["Direction"], lda_pred)
```



```

print("Confusion Matrix:")
print(cm_lda)

# Accuracy
lda_accuracy = accuracy_score(test_data["Direction"], lda_pred)

print("\nAccuracy:", round(lda_accuracy, 4))

# Labeled table
cm_lda_df = pd.DataFrame(
    cm_lda,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_lda_df)

```

Confusion Matrix:

```
[[ 9 34]
 [ 5 56]]
```

Accuracy: 0.625

	Predicted Down	Predicted Up
Actual Down	9	34
Actual Up	5	56

Interpretation

The Linear Discriminant Analysis (LDA) model was trained on observations from 1990–2008 and tested on observations from 2009–2010 using Lag2 as the only predictor.

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	9	34

	Predicted Down	Predicted Up
Actual Up	5	56

The overall test accuracy is **62.5%**.

The LDA model correctly classified 65 of the 104 test observations. It correctly predicted 56 of the 61 Up weeks and 9 of the 43 Down weeks.

The results are identical to those obtained from logistic regression in Part (d). This indicates that both methods produce essentially the same classification boundary when Lag2 is used as the sole predictor.

As with logistic regression, LDA tends to predict Up more often than Down, leading to strong performance on Up weeks but relatively poor performance on Down weeks. Nevertheless, the model achieves a reasonable out-of-sample accuracy of 62.5%, suggesting that Lag2 contains useful information for predicting market direction.

Part (f)

Repeat part (d) using Quadratic Discriminant Analysis (QDA). Compute the confusion matrix and overall accuracy.

```
In [10]: # Part (f): QDA using Lag2

# Training and test predictors
X_train = train_data[["Lag2"]]
X_test = test_data[["Lag2"]]

y_train = train_data["Direction"]

# Fit QDA
qda_model = QDA()
qda_model.fit(X_train, y_train)

# Predictions
qda_pred = qda_model.predict(X_test)

# Confusion matrix
cm_qda = confusion_matrix(test_data["Direction"], qda_pred)
```

```

print("Confusion Matrix:")
print(cm_qda)

# Accuracy
qda_accuracy = accuracy_score(test_data["Direction"], qda_pred)

print("\nAccuracy:", round(qda_accuracy, 4))

# Labeled table
cm_qda_df = pd.DataFrame(
    cm_qda,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_qda_df)

```

Confusion Matrix:

```
[[ 0 43]
 [ 0 61]]
```

Accuracy: 0.5865

	Predicted Down	Predicted Up
Actual Down	0	43
Actual Up	0	61

Interpretation

The Quadratic Discriminant Analysis (QDA) model was trained on observations from 1990–2008 and tested on observations from 2009–2010 using Lag2 as the only predictor.

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	0	43
Actual Up	0	61

The overall test accuracy is **58.7%**.

The QDA model predicted every test observation as **Up**. As a result, it correctly classified all 61 Up weeks but failed to identify any of the 43 Down weeks.

Although the accuracy is moderately high because Up weeks occur more frequently than Down weeks, the model provides no ability to distinguish between the two classes. Compared with logistic regression and LDA (both achieving 62.5% accuracy), QDA performs worse and offers little practical predictive value in this setting.

These results suggest that the more flexible quadratic decision boundary of QDA does not improve classification performance when Lag2 is used as the sole predictor.

Part (g) KNN with K=1

```
In [11]: # Part (g): KNN (K=1) using Lag2

X_train = train_data[["Lag2"]]
X_test = test_data[["Lag2"]]

y_train = train_data["Direction"]
y_test = test_data["Direction"]

# KNN with K=1
knn1 = KNeighborsClassifier(n_neighbors=1)
knn1.fit(X_train, y_train)

knn1_pred = knn1.predict(X_test)

# Confusion matrix
cm_knn1 = confusion_matrix(y_test, knn1_pred)

print("Confusion Matrix:")
print(cm_knn1)

# Accuracy
knn1_acc = accuracy_score(y_test, knn1_pred)

print("\nAccuracy:", round(knn1_acc, 4))
```

```
# Labeled table
cm_knn1_df = pd.DataFrame(
    cm_knn1,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_knn1_df)
```

Confusion Matrix:

```
[[21 22]
 [30 31]]
```

Accuracy: 0.5

	Predicted Down	Predicted Up
Actual Down	21	22
Actual Up	30	31

Part (h)

Repeat part (d) using Naive Bayes. Compute the confusion matrix and overall accuracy.

In [12]: *# Part (h): Naive Bayes using Lag2*

```
# Training and test data
X_train = train_data[["Lag2"]]
X_test = test_data[["Lag2"]]

y_train = train_data["Direction"]

# Fit Naive Bayes
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)

# Predictions
nb_pred = nb_model.predict(X_test)
```

```
# Confusion matrix
cm_nb = confusion_matrix(test_data["Direction"], nb_pred)

print("Confusion Matrix:")
print(cm_nb)

# Accuracy
nb_accuracy = accuracy_score(test_data["Direction"], nb_pred)

print("\nAccuracy:", round(nb_accuracy, 4))

# Labeled table
cm_nb_df = pd.DataFrame(
    cm_nb,
    index=["Actual Down", "Actual Up"],
    columns=["Predicted Down", "Predicted Up"]
)

display(cm_nb_df)
```

Confusion Matrix:

```
[[ 0 43]
 [ 0 61]]
```

Accuracy: 0.5865

	Predicted Down	Predicted Up
Actual Down	0	43
Actual Up	0	61

Interpretation

The Naive Bayes model was trained on observations from 1990–2008 and tested on observations from 2009–2010 using Lag2 as the only predictor.

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	0	43
Actual Up	0	61

The overall test accuracy is **58.7%**.

The Naive Bayes classifier predicted every test observation as **Up**. Consequently, it correctly classified all 61 Up weeks but failed to identify any of the 43 Down weeks.

Although the model achieves moderate accuracy because Up weeks are more common than Down weeks, it provides no ability to distinguish between the two classes. Its performance is identical to that of QDA and worse than that of logistic regression and LDA, which both achieved 62.5% accuracy on the test data.

These results suggest that Naive Bayes does not capture sufficient information from Lag2 alone to effectively discriminate between Up and Down market movements.

Interpretation

The K-Nearest Neighbors (KNN) model with $K = 1$ was trained on observations from 1990–2008 and tested on observations from 2009–2010 using Lag2 as the only predictor.

The confusion matrix is:

	Predicted Down	Predicted Up
Actual Down	21	22
Actual Up	30	31

The overall test accuracy is **50.0%**.

The model correctly classified 21 of the 43 Down weeks and 31 of the 61 Up weeks, resulting in 52 correct predictions out of 104 test observations.

Unlike logistic regression, LDA, QDA, and Naive Bayes, KNN does not exhibit a strong bias toward predicting Up. Instead, it produces a more balanced set of predictions across the two classes. However, this balance comes at the cost of lower overall accuracy.

With an accuracy of only 50%, the KNN model performs no better than random guessing and is the weakest-performing method among those considered. This suggests that using only Lag2 with a highly flexible KNN classifier does not provide reliable predictive power for market direction.

Part (i)

Overall Conclusion

Among the methods considered, Logistic Regression and LDA achieved the best out-of-sample performance, both obtaining a test accuracy of 62.5%.

QDA and Naive Bayes performed worse, each achieving 58.7% accuracy and effectively predicting every observation as Up. KNN with $K = 1$ performed the worst, achieving only 50.0% accuracy.

These results suggest that Lag2 contains some predictive information about future market direction, but the relationship is relatively weak. Simpler linear methods such as Logistic Regression and LDA appear to capture this information more effectively than more flexible methods such as QDA, Naive Bayes, and KNN.

```
In [13]: # Part (j): Experiment with different predictors and K values

predictor_sets = {
    "Lag2": ["Lag2"],
    "Lag1_Lag2": ["Lag1", "Lag2"],
    "Lag1_Lag2_Volume": ["Lag1", "Lag2", "Volume"],
    "Lag1_to_Lag5": ["Lag1", "Lag2", "Lag3", "Lag4", "Lag5"]
}

results = []

for name, predictors in predictor_sets.items():
    X_train = sm.add_constant(train_data[predictors])
    X_test = sm.add_constant(test_data[predictors])
    y_train_binary = (train_data["Direction"] == "Up").astype(int)
```



```
model = sm.Logit(y_train_binary, X_train).fit(dis=False)
probs = model.predict(X_test)
preds = np.where(probs > 0.5, "Up", "Down")

acc = accuracy_score(test_data["Direction"], preds)
cm = confusion_matrix(test_data["Direction"], preds)

results.append(["Logistic Regression", name, acc, cm])

for k in [1, 3, 5, 7, 9, 11]:
    knn = KNeighborsClassifier(n_neighbors=k)
    knn.fit(train_data["Lag2"], train_data["Direction"])
    preds = knn.predict(test_data["Lag2"])

    acc = accuracy_score(test_data["Direction"], preds)
    cm = confusion_matrix(test_data["Direction"], preds)

    results.append([f"KNN K={k}", "Lag2", acc, cm])

for result in results:
    print(result[0], "| Predictors:", result[1])
    print("Accuracy:", round(result[2], 4))
    print(result[3])
    print()
```

Logistic Regression | Predictors: Lag2

Accuracy: 0.625

```
[[ 9 34]
 [ 5 56]]
```

Logistic Regression | Predictors: Lag1_Lag2

Accuracy: 0.5769

```
[[ 7 36]
 [ 8 53]]
```

Logistic Regression | Predictors: Lag1_Lag2_Volume

Accuracy: 0.5288

```
[[27 16]
 [33 28]]
```

Logistic Regression | Predictors: Lag1_to_Lag5

Accuracy: 0.5481

```
[[10 33]
 [14 47]]
```

KNN K=1 | Predictors: Lag2

Accuracy: 0.5

```
[[21 22]
 [30 31]]
```

KNN K=3 | Predictors: Lag2

Accuracy: 0.5577

```
[[16 27]
 [19 42]]
```

KNN K=5 | Predictors: Lag2

Accuracy: 0.5288

```
[[15 28]
 [21 40]]
```

KNN K=7 | Predictors: Lag2

Accuracy: 0.5481

```
[[16 27]
 [20 41]]
```

KNN K=9 | Predictors: Lag2

Accuracy: 0.5481

```
[[17 26]  
 [21 40]]
```

KNN K=11 | Predictors: Lag2

Accuracy: 0.5577

```
[[18 25]  
 [21 40]]
```

Part (j)

Several alternative models were evaluated on the held-out test data. Logistic Regression was fit using:

- Lag2
- Lag1 + Lag2
- Lag1 + Lag2 + Volume
- Lag1–Lag5

KNN models were also evaluated using K = 1, 3, 5, 7, 9, and 11.

Among all models tested, Logistic Regression using Lag2 alone achieved the highest test accuracy (62.5%). LDA using Lag2, evaluated in Part (e), achieved the same accuracy and therefore tied for the best overall performance. Based on the results obtained, the best held-out test performance was achieved by Logistic Regression and LDA using Lag2 as the predictor.

Confusion Matrix (Logistic Regression / LDA):

	Predicted Down	Predicted Up
Actual Down	9	34
Actual Up	5	56

Accuracy = 62.5%

Additional methods were evaluated:

- QDA: 58.7%
- Naive Bayes: 58.7%

- KNN (best accuracy = 55.8% for K=3 and K=11)

Although KNN produced a more balanced set of predictions, its overall accuracy was substantially lower. QDA and Naive Bayes predicted nearly all observations as Up and failed to identify Down weeks effectively.

Additional experiments with alternative predictor combinations did not improve performance beyond Logistic Regression and LDA using Lag2 alone.

Therefore, the best-performing model on the held-out data was Logistic Regression (tied with LDA), using Lag2 as the predictor, achieving an accuracy of 62.5%.

Multi-class Prediction of Obesity Risk

```
In [20]: train = pd.read_csv("obesity-risk/train.csv")
test = pd.read_csv("obesity-risk/test.csv")
sample_submission = pd.read_csv("obesity-risk/sample_submission.csv")
```

```
In [21]: train = pd.read_csv("obesity-risk/train.csv")

print(train.shape)
display(train.head())
print(train.columns.tolist())
```

(20758, 18)

	id	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SM
0	0	Male	24.443011	1.699998	81.669950	yes	yes	2.000000	2.983297	Sometimes	
1	1	Female	18.000000	1.560000	57.000000	yes	yes	2.000000	3.000000	Frequently	
2	2	Female	18.000000	1.711460	50.165754	yes	yes	1.880534	1.411685	Sometimes	
3	3	Female	20.952737	1.710730	131.274851	yes	yes	3.000000	3.000000	Sometimes	
4	4	Male	31.641081	1.914186	93.798055	yes	yes	2.679664	1.971472	Sometimes	

```
['id', 'Gender', 'Age', 'Height', 'Weight', 'family_history_with_overweight', 'FAVC', 'FCVC', 'NCP', 'CAE  
C', 'SMOKE', 'CH20', 'SCC', 'FAF', 'TUE', 'CALC', 'MTRANS', 'NObeyesdad']
```

Data Preparation

In [22]: *# Separate predictors and response*

```
X = train.drop(columns=["id", "NObeyesdad"])
y = train["NObeyesdad"]

X_test_kaggle = test.drop(columns=["id"])

# Identify categorical and numerical variables
categorical_features = X.select_dtypes(include=["object"]).columns.tolist()
numeric_features = X.select_dtypes(include=["int64", "float64"]).columns.tolist()

print("Categorical variables:", categorical_features)
print("Numeric variables:", numeric_features)

print("\nTarget classes:")
print(y.value_counts())
```

```
Categorical variables: ['Gender', 'family_history_with_overweight', 'FAVC', 'CAEC', 'SMOKE', 'SCC', 'CALC',  
'MTRANS']
```

```
Numeric variables: ['Age', 'Height', 'Weight', 'FCVC', 'NCP', 'CH20', 'FAF', 'TUE']
```

Target classes:

NObeyesdad

Obesity_Type_III 4046

Obesity_Type_II 3248

Normal_Weight 3082

Obesity_Type_I 2910

Insufficient_Weight 2523

Overweight_Level_II 2522

Overweight_Level_I 2427

Name: count, dtype: int64

In [23]: *# Train-validation split*

```
X_train, X_val, y_train, y_val = train_test_split(  
    X,
```

```

    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training size:", X_train.shape)
print("Validation size:", X_val.shape)

```

Training size: (16606, 16)
 Validation size: (4152, 16)

Preprocessing Pipeline

```

In [24]: from sklearn.preprocessing import OneHotEncoder, StandardScaler
        from sklearn.compose import ColumnTransformer

        # Preprocess data

        preprocessor = ColumnTransformer(
            transformers=[
                ("num", StandardScaler(), numeric_features),
                ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_features)
            ]
        )

        preprocessor

```

```

Out[24]:
  ▸ ColumnTransformer ⓘ ⓘ
    ▸ num                ▸ cat
      ▸ StandardScaler ⓘ   ▸ OneHotEncoder ⓘ

```

Model 1: Multinomial Logistic Regression

```

In [26]: from sklearn.pipeline import Pipeline
        from sklearn.linear_model import LogisticRegression

```

```
from sklearn.metrics import accuracy_score

log_model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(
        max_iter=5000,
        random_state=42
    ))
])

log_model.fit(X_train, y_train)

log_pred = log_model.predict(X_val)

log_acc = accuracy_score(y_val, log_pred)

print("Logistic Regression Accuracy:", round(log_acc, 4))
```

Logistic Regression Accuracy: 0.8687

The multinomial logistic regression model produced a validation accuracy of 86.87%. This indicates strong baseline performance for predicting the seven obesity-risk classes after scaling numeric variables and one-hot encoding categorical variables.

Model 2: Linear Discriminant Analysis (LDA)

```
In [27]: lda_model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LDA())
])

lda_model.fit(X_train, y_train)

lda_pred = lda_model.predict(X_val)

lda_acc = accuracy_score(y_val, lda_pred)

print("LDA Accuracy:", round(lda_acc, 4))
```

LDA Accuracy: 0.823

The Linear Discriminant Analysis (LDA) model achieved a validation accuracy of 82.30%. While the model performed well, its accuracy was lower than the multinomial logistic regression model, suggesting that the assumption of equal covariance structures across classes may not fully capture the relationships present in the obesity-risk data.

Model 3: Naive Bayes

```
In [28]: from sklearn.naive_bayes import GaussianNB

# Transform data because GaussianNB cannot handle sparse matrices
X_train_nb = preprocessor.fit_transform(X_train)
X_val_nb = preprocessor.transform(X_val)

# Convert sparse matrix to dense if needed
if hasattr(X_train_nb, "toarray"):
    X_train_nb = X_train_nb.toarray()
    X_val_nb = X_val_nb.toarray()

nb_model = GaussianNB()

nb_model.fit(X_train_nb, y_train)

nb_pred = nb_model.predict(X_val_nb)

nb_acc = accuracy_score(y_val, nb_pred)

print("Naive Bayes Accuracy:", round(nb_acc, 4))
```

Naive Bayes Accuracy: 0.586

The Gaussian Naive Bayes model achieved a validation accuracy of 58.60%, substantially lower than both Logistic Regression and LDA. This result suggests that the strong independence assumptions made by Naive Bayes are not appropriate for this dataset, where many lifestyle and health-related predictors are likely correlated.

Model 4: Support Vector Machine (SVM)

```
In [29]: from sklearn.svm import SVC
```



```
svm_model = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", SVC(
        kernel="rbf",
        C=1.0,
        gamma="scale",
        random_state=42
    ))
])

svm_model.fit(X_train, y_train)

svm_pred = svm_model.predict(X_val)

svm_acc = accuracy_score(y_val, svm_pred)

print("SVM Accuracy:", round(svm_acc, 4))
```

SVM Accuracy: 0.881

The Support Vector Machine (SVM) achieved the highest validation accuracy of 88.10%, outperforming Logistic Regression, LDA, and Naive Bayes. The nonlinear decision boundaries created by the radial basis function (RBF) kernel appear well-suited to the complex relationships among demographic, dietary, and lifestyle variables associated with obesity risk. Consequently, SVM was selected as the best-performing classification model.

```
In [30]: log_model.fit(X, y)

log_test_pred = log_model.predict(X_test_kaggle)

submission_log = pd.DataFrame({
    "id": test["id"],
    "NObeyesdad": log_test_pred
})

submission_log.to_csv("submission_logistic.csv", index=False)

submission_log.head()
```

Out [30]:

	id	NObeyesdad
0	20758	Obesity_Type_II
1	20759	Overweight_Level_I
2	20760	Obesity_Type_III
3	20761	Obesity_Type_I
4	20762	Obesity_Type_III

In [31]:

```
lda_model.fit(X, y)

lda_test_pred = lda_model.predict(X_test_kaggle)

submission_lda = pd.DataFrame({
    "id": test["id"],
    "NObeyesdad": lda_test_pred
})

submission_lda.to_csv("submission_lda.csv", index=False)

submission_lda.head()
```

Out [31]:

	id	NObeyesdad
0	20758	Obesity_Type_II
1	20759	Overweight_Level_I
2	20760	Obesity_Type_III
3	20761	Obesity_Type_II
4	20762	Obesity_Type_III

In [32]:

```
# Refit Naive Bayes on the full training data

X_full_nb = preprocessor.fit_transform(X)
X_test_nb = preprocessor.transform(X_test_kaggle)
```

```

if hasattr(X_full_nb, "toarray"):
    X_full_nb = X_full_nb.toarray()
    X_test_nb = X_test_nb.toarray()

nb_model = GaussianNB()
nb_model.fit(X_full_nb, y)

nb_test_pred = nb_model.predict(X_test_nb)

submission_nb = pd.DataFrame({
    "id": test["id"],
    "NObeyesdad": nb_test_pred
})

submission_nb.to_csv("submission_nb.csv", index=False)

submission_nb.head()

```

Out [32]:

	id	NObeyesdad
0	20758	Obesity_Type_II
1	20759	Obesity_Type_I
2	20760	Obesity_Type_III
3	20761	Obesity_Type_II
4	20762	Obesity_Type_III

In [33]: *# Refit SVM on the full training data*

```

svm_model.fit(X, y)

svm_test_pred = svm_model.predict(X_test_kaggle)

submission_svm = pd.DataFrame({
    "id": test["id"],
    "NObeyesdad": svm_test_pred
})

submission_svm.to_csv("submission_svm.csv", index=False)

```

```
submission_svm.head()
```

Out [33]:

	id	NObeyesdad
0	20758	Obesity_Type_II
1	20759	Overweight_Level_I
2	20760	Obesity_Type_III
3	20761	Obesity_Type_I
4	20762	Obesity_Type_III

Kaggle Submission Results

All four classification models were submitted to the Kaggle Multi-Class Prediction of Obesity Risk competition.

Model	Public Kaggle Score
Support Vector Machine (SVM)	0.88005
Logistic Regression	0.86416
Linear Discriminant Analysis (LDA)	0.82153
Naive Bayes	0.58995

The Kaggle leaderboard results were consistent with the validation results obtained during model development. The Support Vector Machine achieved the highest predictive performance and ranked as the best overall model. Logistic Regression also performed strongly, while LDA achieved moderate performance. Naive Bayes produced the weakest results, suggesting that the independence assumption among predictors was not appropriate for this dataset.

Regularized Logistic Regression

```
In [35]: from sklearn.linear_model import LogisticRegression

ridge_log = Pipeline([
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(
        C=1.0,
        max_iter=5000,
        random_state=42
    ))
])

ridge_log.fit(X_train, y_train)

ridge_pred = ridge_log.predict(X_val)

ridge_acc = accuracy_score(y_val, ridge_pred)

print("Regularized Logistic Regression Accuracy:", round(ridge_acc, 4))
```

Regularized Logistic Regression Accuracy: 0.8687

The regularized logistic regression model achieved an accuracy of 86.87%, identical to the standard logistic regression model. This indicates that the original model was already well-generalized and not substantially affected by overfitting. The use of L2 regularization provides additional protection against overfitting while maintaining predictive performance.

References

Ghojogh, B., & Crowley, M. (2019). Linear and quadratic discriminant analysis: Tutorial. *arXiv*. <https://arxiv.org/abs/1906.02590>

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer.

Kumar, A., Sharma, S., & Gupta, R. (2023). Obesity disease risk prediction using machine learning. *International Journal of Information Technology*, 15, 1–12. <https://doi.org/10.1007/s41060-023-00491-9>

In []: